ACLs

L2 Switch (SW1)

```
Extended IP access list NO_SSH
    10 permit tcp host 192.168.5.201 host 192.168.5.205 eq 22
Extended IP access list NO_SSH2
Standard IP access list NO_SSH3
    10 permit 192.168.5.200 0.0.0.7
```

L3 Switch (SW2)

```
Extended IP access list VLAN10
    10 deny ip 192.168.5.0 0.0.0.63 192.168.5.64 0.0.0.63
    20 deny ip 192.168.5.0 0.0.0.63 192.168.5.128 0.0.0.63
    30 deny ip 192.168.5.0 0.0.0.63 192.168.5.200 0.0.0.7
    40 permit udp any any eq bootps
    50 permit udp any eq bootpc any eq bootps
    60 permit ip any any
Extended IP access list VLAN20
    10 deny ip 192.168.5.64 0.0.0.63 192.168.5.0 0.0.0.63
    20 deny ip 192.168.5.64 0.0.0.63 192.168.5.128 0.0.0.63
    30 deny ip 192.168.5.64 0.0.0.63 192.168.5.200 0.0.0.7
    40 permit udp any any eq bootps
    50 permit udp any eq bootpc any eq bootps
    60 permit ip any any
Extended IP access list VLAN30
    10 deny ip 192.168.5.128 0.0.0.63 192.168.5.0 0.0.0.63
    20 deny ip 192.168.5.128 0.0.0.63 192.168.5.64 0.0.0.63
    30 deny ip 192.168.5.128 0.0.0.63 192.168.5.200 0.0.0.7
    40 permit udp any any eq bootps
    50 permit udp any eq bootpc any eq bootps
    60 permit ip any any
Standard IP access list NO_SSH
    10 permit 192.168.5.200 0.0.0.7
```

Router (R1)

```
Extended IP access list WAN_IN
    10 permit tcp any any eq www
    20 permit tcp any any eq 443
    30 permit udp any any eq domain
    40 permit tcp 192.168.5.200 0.0.0.7 any eq 22
```

DHCP server Configuration

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
VLAN 30	192.168.5.129	8.8.8.8	192.168.5.130	255.255.255.192	62	0.0.0.0	0.0.0.0
VLAN 20	192.168.5.65	8.8.8.8	192.168.5.66	255.255.255.192	62	0.0.0.0	0.0.0.0
VLAN 10	192.168.5.2	8.8.8.8	192.168.5.3	255.255.255.192	61	0.0.0.0	0.0.0.0
serverPool	0.0.0.0	0.0.0.0	192.168.5.192	255.255.255.248	512	0.0.0.0	0.0.0.0

Router interfaces

```
interface GigabitEthernet0/0
 ip address 192.168.5.209 255.255.255.252
 ip access-group WAN_IN in
 duplex auto
 speed auto
!
interface GigabitEthernet0/1
 no ip address
 duplex auto
 speed auto
 shutdown
!
interface GigabitEthernet0/2
 no ip address
 duplex auto
 speed auto
 shutdown
!
interface Vlan1
 no ip address
 shutdown
!
```

L2 Switch

```
.
interface FastEthernet0/1
 description Used for Human resource
 switchport access vlan 30
 ip access-group NO_SSH in
 spanning-tree portfast
 spanning-tree bpduguard enable
!
interface FastEthernet0/2
 description Used for Human resource
 switchport access vlan 30
 ip access-group NO_SSH in
 spanning-tree portfast
 spanning-tree bpduguard enable
!
interface FastEthernet0/3
 description Used for Human resource
 switchport access vlan 30
 ip access-group NO_SSH in
 spanning-tree portfast
 spanning-tree bpduguard enable
!
interface FastEthernet0/4
 description Used for Human resource
 switchport access vlan 30
 ip access-group NO_SSH in
 spanning-tree portfast
 spanning-tree bpduguard enable
!
interface FastEthernet0/5
 description Used for Engineering
 switchport access vlan 20
 ip access-group NO_SSH in
 spanning-tree portfast
 spanning-tree bpduguard enable
!
interface FastEthernet0/6
 description Used for Engineering
 switchport access vlan 20
 ip access-group NO_SSH in
 spanning-tree portfast
 spanning-tree bpduguard enable
!
interface FastEthernet0/7
 description Used for Engineering
 switchport access vlan 20
 ip access-group NO_SSH in
 spanning-tree portfast
 spanning-tree bpduguard enable
!
interface FastEthernet0/8
 description Used for Engineering
 switchport access vlan 20
 ip access-group NO_SSH in
 spanning-tree portfast
 spanning-tree bpduguard enable
!
interface FastEthernet0/9
 description Used for Engineering
 switchport access vlan 20
 ip access-group NO_SSH in
 spanning-tree portfast
 spanning-tree bpduguard enable
 --More--
```

```
VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/11, Fa0/12, Fa0/13, Fa0/14
                                                Fa0/15, Fa0/16, Fa0/17, Fa0/18
                                                Fa0/19, Fa0/21, Gig0/1, Gig0/2
10   Finance                          active
20   Engineering                      active    Fa0/5, Fa0/6, Fa0/7, Fa0/8
                                                Fa0/9, Fa0/10
30   HR                               active    Fa0/1, Fa0/2, Fa0/3, Fa0/4
40   Servers                          active    Fa0/22, Fa0/23
99   Network_Management               active    Fa0/20
```

L3 Switch

```
VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/11, Fa0/12, Fa0/13, Fa0/1
                                                Fa0/15, Fa0/16, Fa0/17, Fa0/1
                                                Fa0/19, Fa0/20, Fa0/21, Fa0/2
                                                Fa0/23, Gig0/1, Gig0/2
10   Finance                          active    Fa0/2, Fa0/3, Fa0/4, Fa0/5
                                                Fa0/6, Fa0/7, Fa0/8, Fa0/9
                                                Fa0/10
20   Engineering                      active
30   HR                               active
40   Server                           active
99   Network_Management               active

Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/1        192.168.5.210   YES manual up                    up
FastEthernet0/2        unassigned      YES unset  up                    up
FastEthernet0/3        unassigned      YES unset  up                    up
FastEthernet0/4        unassigned      YES unset  up                    up
FastEthernet0/5        unassigned      YES unset  up                    up
FastEthernet0/6        unassigned      YES unset  up                    up
FastEthernet0/7        unassigned      YES unset  up                    up
FastEthernet0/8        unassigned      YES unset  down                  down
FastEthernet0/9        unassigned      YES unset  down                  down
FastEthernet0/10       unassigned      YES unset  down                  down
FastEthernet0/11       unassigned      YES unset  down                  down
FastEthernet0/12       unassigned      YES unset  down                  down
FastEthernet0/13       unassigned      YES unset  down                  down
FastEthernet0/14       unassigned      YES unset  down                  down
FastEthernet0/15       unassigned      YES unset  down                  down
FastEthernet0/16       unassigned      YES unset  down                  down
FastEthernet0/17       unassigned      YES unset  down                  down
FastEthernet0/18       unassigned      YES unset  down                  down
FastEthernet0/19       unassigned      YES unset  down                  down
FastEthernet0/20       unassigned      YES unset  down                  down
FastEthernet0/21       unassigned      YES unset  down                  down
FastEthernet0/22       unassigned      YES unset  down                  down
FastEthernet0/23       unassigned      YES unset  down                  down
FastEthernet0/24       unassigned      YES unset  up                    up
GigabitEthernet0/1     unassigned      YES unset  down                  down
GigabitEthernet0/2     unassigned      YES unset  down                  down
Vlan1                  unassigned      YES unset  administratively down down
Vlan10                 192.168.5.2     YES manual up                    up
Vlan20                 192.168.5.65    YES manual up                    up
Vlan30                 192.168.5.129   YES manual up                    up
Vlan40                 192.168.5.193   YES manual up                    up
Vlan99                 192.168.5.206   YES manual up                    up
```

```
interface FastEthernet0/1
 no switchport
 ip address 192.168.5.210 255.255.255.252
 duplex auto
 speed auto
!
interface FastEthernet0/2
 switchport access vlan 10
 spanning-tree portfast
 spanning-tree bpduguard enable
!
interface FastEthernet0/3
 switchport access vlan 10
 spanning-tree portfast
 spanning-tree bpduguard enable
!
interface FastEthernet0/4
 switchport access vlan 10
 spanning-tree portfast
 spanning-tree bpduguard enable
!
interface FastEthernet0/5
 switchport access vlan 10
 spanning-tree portfast
 spanning-tree bpduguard enable
!
interface FastEthernet0/6
 switchport access vlan 10
 spanning-tree portfast
 spanning-tree bpduguard enable
!
interface FastEthernet0/7
 switchport access vlan 10
 spanning-tree portfast
 spanning-tree bpduguard enable
!
interface FastEthernet0/8
 switchport access vlan 10
 spanning-tree portfast
 spanning-tree bpduguard enable
!
interface FastEthernet0/9
 switchport access vlan 10
 spanning-tree portfast
 spanning-tree bpduguard enable
!
interface FastEthernet0/10
 switchport access vlan 10
 spanning-tree portfast
 spanning-tree bpduguard enable
!
```

SSH configuration L2 switch

```
ip ssh version 2
ip domain-name ciscoswitch123.com
!
username admin privilege 1 password 0 cisco123
!
```

```
line vty 0 4
 access-class NO_SSH3 in
 login local
 transport input ssh
line vty 5 15
 access-class NO_SSH3 in
 login local
 transport input ssh
```

L3 switch

```
username admin password 0 cisco123
!
!
!
!
!
!
!
!
!
ip ssh version 2
ip domain-name ciscoswitch.com
!
!
line vty 0 4
 access-class NO_SSH in
 login local
 transport input ssh
line vty 5 15
 access-class NO_SSH in
 login local
 transport input ssh
!
```

Router

```
!
username admin password 0 cisco123
!
!
license udi pid CISCO2911/K9 sn FTX1524WCGM-
!
!
!
!
!
!
!
!
!
!
ip ssh version 2
ip domain-name cisco123
!
!
line vty 0 4
 login local
 transport input ssh
line vty 5 15
 login local
 transport input ssh
!
```

Static routing

Router

```
!
ip classless
ip route 192.168.5.0 255.255.255.192 192.168.5.202
ip route 192.168.5.64 255.255.255.192 192.168.5.202
ip route 192.168.5.128 255.255.255.192 192.168.5.202
ip route 192.168.5.192 255.255.255.248 192.168.5.202
ip route 192.168.5.205 255.255.255.255 192.168.5.202
ip route 192.168.5.0 255.255.255.192 192.168.5.210
ip route 192.168.5.64 255.255.255.192 192.168.5.210
ip route 192.168.5.128 255.255.255.192 192.168.5.210
ip route 192.168.5.192 255.255.255.248 192.168.5.210
ip route 192.168.5.200 255.255.255.248 192.168.5.210
!
```

L3 switch

```
ip ---------
ip route 0.0.0.0 0.0.0.0 192.168.5.201
!
```